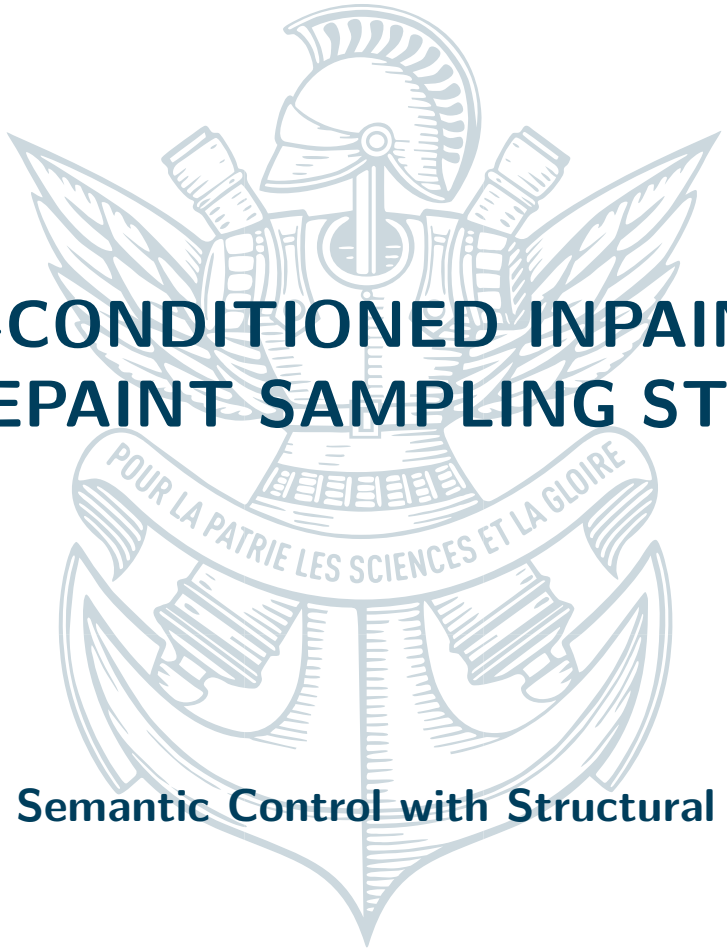


TEXT-CONDITIONED INPAINTING WITH REPAINT SAMPLING STRATEGY

Combining Semantic Control with Structural Coherence

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Abstract

Image inpainting is the task of filling in missing or damaged parts of an image in a visually plausible manner. While traditional approaches focus on maintaining structural and textural consistency, they often lack semantic control over the generated content. We present an approach that combines text-guided generation using GLIDE's classifier-free guidance with RePaint's sampling strategy for high-quality inpainting. Though newer models like Stable Diffusion represent the current state of the art, our work explores how better sampling strategies can enhance pixel-space diffusion models. We discuss the challenges of adapting such strategies to latent-space models and provide insights for future work. Our experimental results demonstrate that the combined approach performs slightly better than GLIDE's standard inpainting method. We evaluate our approach on multiple datasets including Places365, ImageNet-Micro, and MS-COCO, using both standard metrics (FID, LPIPS) and human evaluation.

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1 Introduction

Image inpainting is the task of filling in missing or damaged parts of an image in a visually plausible manner. This task has numerous applications, including photo restoration, object removal, and creative image editing. Traditional inpainting approaches focus on maintaining structural and textural consistency but often lack semantic control over the generated content.

Recent years have seen rapid advancement in generative models for image synthesis and editing. While newer text-to-image models like Stable Diffusion [6] currently represent the state-of-the-art in this domain, there remains value in exploring improvements to earlier models such as GLIDE [5], particularly when combined with better sampling strategies.

In this work, we present an exploratory study that combines GLIDE's text-conditional diffusion model with RePaint's sampling strategy [4] for image inpainting. It is important to acknowledge that GLIDE, while groundbreaking at its release in 2021, has been surpassed by newer models in terms of generation quality and capabilities. However, this combination offers an interesting case study in how sampling strategies can improve diffusion model performance.

A critical distinction that affects our approach is that both GLIDE and RePaint operate in pixel space, directly modeling RGB pixel values, unlike newer models such as Stable Diffusion which operate in a compressed latent space. This distinction has several implications:

- **Computational Efficiency:** Pixel-space diffusion is more computationally expensive, especially at higher resolutions, as the model must process each pixel individually.
- **Detail Preservation:** Pixel-space models potentially preserve finer details but at the cost of slower sampling.
- **Sampling Strategy Compatibility:** RePaint's strategy is specifically designed for pixel-space diffusion, making it directly applicable to GLIDE but requiring significant adaptation for latent-space models.

If we were to apply RePaint's sampling strategy to latent-space models like Stable Diffusion, several challenges would emerge. The masking process would need to be translated to latent space, where there isn't a one-to-one correspondence between spatial locations in pixel space. Additionally, the concept of "known regions" becomes more complex when working with abstract latent representations rather than direct pixel values.

Our contribution is an integration of GLIDE's text-conditioning capabilities with RePaint's sampling strategy to create a text-conditioned inpainting system that improves upon GLIDE's standard results. This work provides insights into how better sampling strategies can enhance diffusion model performance, potentially informing future improvements in more advanced models.

- **Semantic Control:** Text-guided generation using GLIDE's classifier-free guidance, which allows for control over the semantic content.
- **Structural Coherence:** RePaint's sampling strategy, which ensures better preservation of known image regions and coherence in transitions between known and generated content.

This approach gives users improved control over the content they wish to generate while maintaining the visual quality of the inpainted image, within the limitations of the underlying GLIDE model.

2 Denoising Diffusion Probabilistic Models

2.1 Foundation of Our Approach

Both GLIDE and RePaint are built upon Denoising Diffusion Probabilistic Models (DDPMs) [2], which have emerged as state-of-the-art generative models for image synthesis. DDPMs define a forward Markov process that gradually adds Gaussian noise to an image, and a reverse process that removes noise to generate new samples.

DDPMs are particularly well-suited for inpainting tasks for several reasons:

- **High-quality generation:** DDPMs produce more realistic and diverse images than GANs or VAEs
- **Flexible conditioning:** They can be easily conditioned on various inputs (text, masks, etc.)
- **Progressive generation:** The step-by-step denoising process allows for interventions at each timestep

2.2 Mathematical Formulation

Formally, the forward diffusion process is defined as:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t\mathbf{I}) \tag{1}$$

where $\beta_t \in [0, 1]$ is a variance schedule that controls the noise added at each step. The forward process can also be expressed in terms of x_0 directly:

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)\mathbf{I}) \tag{2}$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$.

The reverse process is a learned Gaussian transition that gradually removes noise:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \tag{3}$$

The model is trained to predict the noise added at each step, allowing it to reverse the diffusion process and generate new samples. The training objective is to minimize the variational bound on negative log-likelihood:

$$\mathcal{L} = \mathbb{E}_{x_0,\epsilon,t}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\right] \tag{4}$$

where ϵ_θ is a neural network (typically a U-Net) that predicts the noise added at step t.

The sampling algorithm for DDPMs is shown in Algorithm 1.

Algorithm 1 Denoising Diffusion Sampling

1: $x_T \sim \mathcal{N}(0, \mathbf{I})$
2: **for** $t = T, T-1, \ldots, 1$ **do**
3: $\quad z \sim \mathcal{N}(0, \mathbf{I})$ if $t > 1$, else $z = 0$
4: $\quad x_{t-1} = \frac{1}{\sqrt{1-\beta_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right) + \sigma_t z$
5: **end for**
6: **return** x_0

3 Guided Diffusion

3.1 Classifier-Free Guidance

Classifier-Free Guidance (CFG) is a technique introduced to control the generation process of diffusion models without requiring a separate classifier. CFG works by training a conditional model that can also be run unconditionally. During sampling, the model's output is adjusted by interpolating between conditional and unconditional predictions:

$$\hat{\epsilon}_\theta(x_t, t, y) = \epsilon_\theta(x_t, t) + w \cdot (\epsilon_\theta(x_t, t, y) - \epsilon_\theta(x_t, t)) \tag{5}$$

where w is a guidance scale that controls the strength of the conditioning. This approach allows an appropriate tradeoff between sample quality and condition adherence.

3.2 GLIDE Text-Guided Classifier-Free Guidance

GLIDE (Guided Language to Image Diffusion for Generation and Editing) [5] uses classifier-free guidance to condition diffusion models on text. The architecture consists of:

- A text encoder using a transformer architecture to process text prompts
- The text embedding conditioning the U-Net through cross-attention mechanisms
- A diffusion model trained on both conditional (with text) and unconditional examples

For inpainting tasks, GLIDE takes additional inputs:

- The original image with masked regions
- A binary mask indicating which pixels to inpaint

GLIDE's inpainting model is explicitly trained for the inpainting task, with the ability to preserve the known regions of the image while generating coherent content for the masked regions. This makes it particularly effective for controlled image editing.

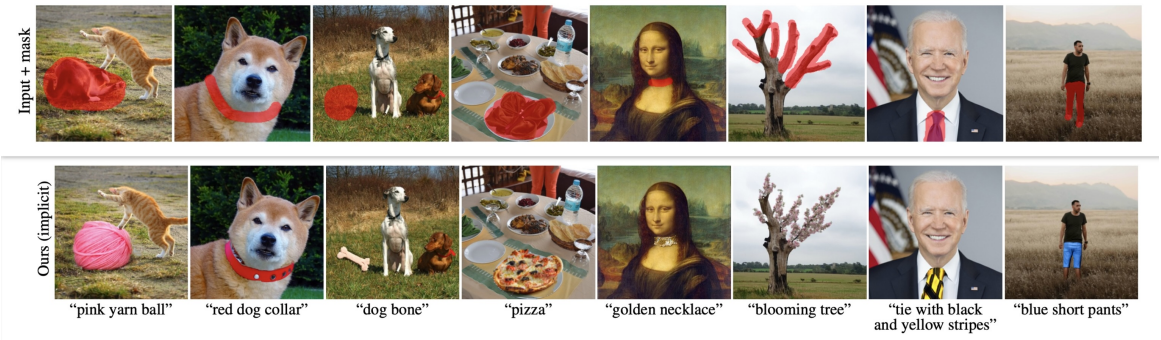


Figure 1: Inpainted samples using GLIDE's text-conditional diffusion.

4 RePaint Sampling Strategy

RePaint [4] improves diffusion-based inpainting through an innovative sampling strategy. Its key innovation is in how it handles the known and unknown regions of the image during the sampling process.

The main steps of RePaint's sampling strategy are:

1. Start from pure noise x_T
2. For each denoising step t:
   - Apply one denoising step to get $\hat{x}_{t-1}$
   - Replace known regions with appropriately noised ground truth:

$$x_{t-1} = M \odot \text{GaussianNoise}(x_0, t-1) + (1-M) \odot \hat{x}_{t-1} \tag{6}$$

3. Implement "resampling" by occasionally increasing t (going back in time)

The resampling approach, where the process occasionally jumps forward in the diffusion process during inference, is crucial for maintaining coherence between the generated content and the known image regions. This approach allows the model to "reconsider" its generation, leading to better inpainting results.

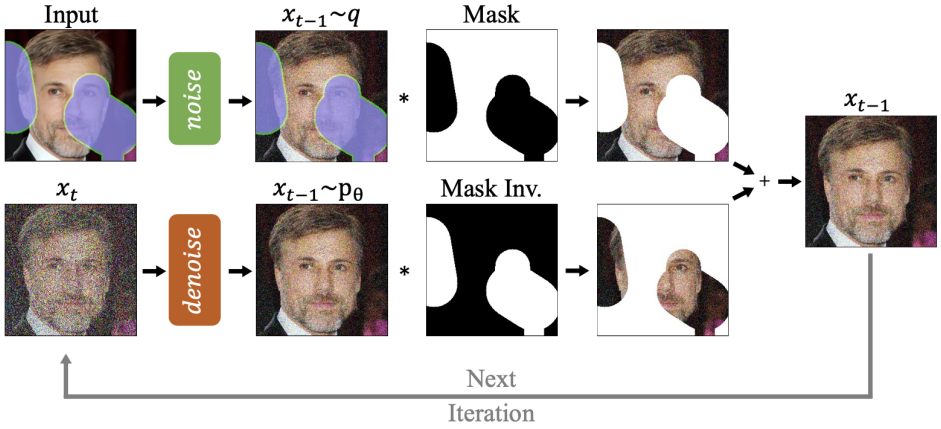


Figure 2: Illustration of the RePaint sampling strategy. At each timestep, known regions (defined by mask M) are replaced with appropriately noised ground truth, while unknown regions are generated by the diffusion model. The schedule includes occasional "jumps" forward in time to improve coherence.

5 Our Approach: Enabling Text Guidance with RePaint Sampling

5.1 Overview

We combine GLIDE's text conditioning with RePaint's sampling strategy to achieve both semantic control and better structural coherence in the inpainting process. Our approach leverages the strengths of both methods:

- GLIDE provides text-guided generation capabilities
- RePaint offers improved handling of known and unknown image regions

The key challenge in this integration is adapting RePaint's sampling strategy to work with GLIDE's classifier-free guidance approach.

5.2 Algorithm

Our text-guided RePaint sampling algorithm is presented in Algorithm 2.

Algorithm 2 Text-Guided RePaint Sampling

Input: Original image x_0, mask M, text prompt y, guidance scale w
Output: Inpainted image
$x_T \sim \mathcal{N}(0, \mathbf{I})$
Initialize text embedding c_y from prompt y
Initialize empty prompt embedding $c_\emptyset$
Define time schedule with jumps $T = \{T, T-1, ..., T-j, ..., T-j+k, ..., 1\}$
for $t \in T$ **do**
  if t is a reverse step **then**
    Predict noise: $\epsilon_\emptyset = \epsilon_\theta(x_t, t)$ and $\epsilon_y = \epsilon_\theta(x_t, t, c_y)$
    Apply guidance: $\hat{\epsilon} = \epsilon_\emptyset + w(\epsilon_y - \epsilon_\emptyset)$
    Compute $\hat{x}_{t-1}$ using $\hat{\epsilon}$
    Get noised known region: $x_{known} = \sqrt{\bar{\alpha}_{t-1}} x_0 + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon$ where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$
    Combine: $x_{t-1} = M \odot x_{known} + (1 - M) \odot \hat{x}_{t-1}$
  else
    Apply forward diffusion step from x_t to x_{t+1}
  end if
end for
return x_0

5.3 Key Innovations

Our approach includes the following innovations:

- **Improved GLIDE's inpainting quality:** The RePaint sampling strategy enhances GLIDE's ability to generate coherent inpainting results.
- **Enabled text guidance at each denoising step with RePaint sampling:** We apply classifier-free guidance at each step of the RePaint sampling process, ensuring that the text conditioning remains effective throughout.
- **Diffusion Schedule optimization:** We developed efficient jumping patterns that balance text-alignment, sample quality, and inference time. Our jumping strategy is designed to provide sufficient resampling while keeping computational costs manageable.

5.4 Challenges in Adapting RePaint to Latent Diffusion Models

Adapting RePaint's sampling strategy to latent diffusion models like Stable Diffusion presents significant challenges. One might consider a naive approach of cycling between latent and pixel space at each diffusion step, i.e. decoding to pixel space, applying RePaint operations, then re-encoding to latent space. However, this approach would face several critical issues:

- **Compounding Encoder/Decoder Errors:** VAEs are lossy compressors, and each encode-decode cycle would introduce artifacts. These errors would accumulate across many diffusion steps, potentially leading to significant degradation in the final result.
- **Statistical Domain Mismatch:** The diffusion process in Stable Diffusion is calibrated specifically for its latent space statistics. The noise schedule assumes certain properties of the latent distribution. Mixing pixel-space operations with latent diffusion would break these statistical assumptions.
- **Computational Inefficiency:** The primary advantage of latent diffusion is reduced computational cost. Encoding/decoding at every step would negate this advantage and likely be significantly slower than even pixel-space diffusion.

- **Mask Projection Issues:** The VAE's downsampling means that one latent "pixel" affects/is affected by multiple pixel-space pixels. This makes accurate mask boundaries difficult to maintain through encode-decode cycles.

A more efficient approach would operate directly in latent space:

- **Latent-Space Masking:** Project the binary mask from pixel space to latent space, accounting for downsampling and potentially applying special handling for mask boundaries due to the non-local nature of latent representations.
- **Latent Conditioning:** Adapt RePaint's conditioning step to work with latent representations, applying noising directly to the encoded latent of known regions.
- **Preserve Diffusion Process:** Implement backward jumps directly in latent space, maintaining the statistical properties expected by the UNet.
- **Boundary Treatment:** Special handling for transitions between known and unknown regions, as boundaries in latent space are more diffuse than in pixel space.

This remains an open research direction that could significantly improve the inpainting capabilities of state-of-the-art latent diffusion models.

6 Evaluation Methodology

6.1 Datasets and Processing

We used the following common recognition datasets:

- **Places365 Train:** Training set from the Places dataset [10] of common scenes.
- **Imagenet-Micro:** A small subset of the ImageNet dataset [1], scraped from the web, that is publicly available on Github.
- **MS-COCO Subset:** A subset of the MS-COCO dataset [3], which we used as a benchmark for text-to-image generation.

All images were resized to a resolution of 64×64 pixels for our experiments, and later upscaled to 256×256.

6.2 Experimental Setup

- **Sample Sizes:** At least 640 images are typically required for reliable FID results, so despite our hardware constraints, we used 640 images scaled to 64×64 resolution to compare each model with each mask.
- **Hardware:** We used a combination of a Télécom Paris shared GPU cluster with an A100, as well as Google Colab with a T4.
- **Libraries:** We used PyTorch to implement our models, drawing from the publicly available implementations of GLIDE and RePaint to make our mixed model. We used the publicly available FID [7] and LPIPS [9] libraries for evaluation.
- **Models:** We based our work on Open AI's GLIDE models, with publicly available weights. We work with three variations of the model, one using the GLIDE Inpainting model, one using the base GLIDE model (not finetuned for inpainting), but with RePaint sampling, and one using GLIDE Inpainting with RePaint sampling. Further details are provided in the results section.
- **Hyperparameters:** Inspired by the original RePaint paper, we started with 250 diffusion steps. We experimented with alternative numbers of diffusion steps, and found 250 to be a reasonable amount to see a difference between standard GLIDE sampling and RePaint.

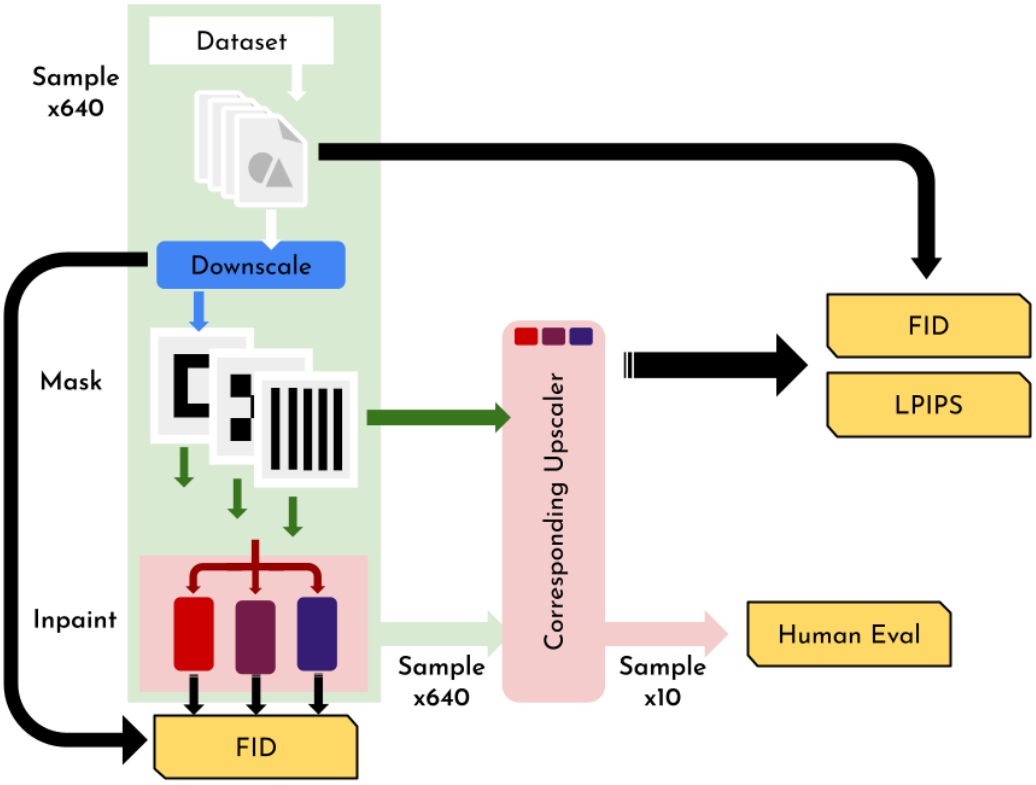


Figure 3: An overview of our evaluation method. Due to time and hardware constraints, we had to reduce the number of evaluation samples. Dark red corresponds to standard GLIDE inpainting, blue corresponds to RePaint sampling with base GLIDE Text2ImUnet model, and Maroon corresponds to our best combined approach.

6.3 Sampling Parameters

A key development in the RePaint paper was a resampling strategy they referred to as jumping. Their jumping strategy effectively caused what they called "250 diffusion steps" to perform 4790 iterations, as they looped forward and backward through the diffusion steps repeatedly. For our evaluation, we use the same definition of $T = 250$ diffusion steps.

We used two sets of modified jumping approach with our RePaint based models. For the un-finetuned RePaint based model, we gave it additional resampling steps to make up for the lack of finetuning. Thus, we chose a jump size of 10, and 5 jump samples. For our model, we restricted it to resampling with a jump length of 10 and 6 jump samples. These parameters were chosen to balance computational efficiency with inpainting quality.

7 Results

For our evaluation, we tested with two baseline models, and our improved combination model. We used the same sampling parameters for the models as described above. The standard GLIDE model finetuned for inpainting, referred to here as *GLIDE-InPaint*, is used as specified by the code provided by OpenAI. We used our patching code to add RePaint sampling to an unfinetuned version of the GLIDE model, which we refer to as *GLIDE-*

Base w/ RePaint. Finally, our chosen method, using the finetuned model with the RePaint sampling strategy, is referred to as *GLIDE-Inpaint w/ RePaint*.

7.1 Timing Analysis

We compared the computational efficiency of our approach with standard GLIDE inpainting. Table 1 shows the performance of different models with and without PyTorch compilation. Timing results with compilation reflect something more similar to the expected processing time if this model was used in a real-world context. For our demo application, we also use PyTorch compilation to increase the responsiveness. We get almost a 2x speedup with compilation, which is much larger than the around 20% speedup often reported on the internet as of now (March 2025).

We compare the time taken per iteration and the overall time to process a batch with the repaint sampling strategy compared to the standard DDPM strategy. Clearly, the repaint sampling strategy takes around half as much time per iteration compared with standard DDPM. An interesting further area of strategy would be comparing the results of additional step cycles as performed by repaint with additional diffusion steps. Based on the original RePaint paper, benefits seem to saturate with cycles of size 10 (meaning 10 backward steps are followed by 10 forward steps), but the scheduler actually provided with RePaint performs additional cycling at every step. Since these weights are already on the GPU, the performance cost of doing these extra steps can be negligible, except that it adds 20x the iterations. However, these additional iterations do seem to improve image quality, often leading to a more cohesive image (compare, for example, in the qualitative results (4), how our method seems to fill in objects that better integrate the prompt and context of the unmasked image, like the matching in era between the vehicles generated with the trolley in the third set).

Model	Uncompiled		PyTorch Compiled	
	Seconds/Iteration	Time per Batch	Seconds/Iteration	Time per Batch
GLIDE-Inpaint	0.6135	153	0.3448	86
GLIDE-Base w/ RePaint	0.3484	657	0.1953	369
GLIDE-Inpaint w/ RePaint (ours)	0.3922	357	0.2203	200

Table 1: Performance comparison of different models with and without PyTorch compilation. Time per batch is in seconds.

7.2 Quantitative Results

7.2.1 Downscaled Inpainting

Model	Mean FID Score Per Mask						
	ex64	genhalf	sr64	thick	thin	vs64	**Mean FID Score**
GLIDE-Inpaint	76.272	48.985	160.327	8.517	56.902	77.633	71.439
GLIDE-Base w/ RePaint	79.628	61.083	157.157	13.810	100.694	113.434	87.634
GLIDE-Inpaint w/ RePaint (ours)	77.035	49.354	162.216	8.774	55.171	88.525	73.512

Table 2: Model performance comparison across different masks.

In Table 2, we have the FID score ([7]) results compared across models and masks, averaged over each dataset. We downscaled 640 images for each mask from each dataset, leading to a total of 15360 images sampled and evaluated for this metric. The full, unaveraged, results are shown in C. In addition, links to zipped versions of all of our generated images are available in the appendix. Unfortunately, the RePaint strategy on the un-finetuned model (GLIDE-Base w/ Repaint) is unable to come close to the finetuned model, at least in the

restricted version of the sampling strategy we were forced to use for performance reasons. However, with even fewer iterations the GLIDE-Inpaint w/ RePaint strategy was able to actually come close to GLIDE-Inpaint, even though we were severely limiting the sampling strategy by not using it as intended. Nevertheless, the FID score on all masks is quite high, because the task is difficult, and we gave all of the models a relatively low number of diffusion steps compared to what is often required for optimal results for diffusion models. In addition, FID may not be the best scoring metric for such small images as we are forced to use by the publicly available GLIDE model, motivating the next subsection.

7.2.2 Upscaled Inpainting

In addition to comparing model performance on the downscaled version of the images, we also evaluated a limited subset of the images generated previously with an upscaling task. For this section, because we had higher resolution images, the processing time was longer and more expensive, thus we selected only 160 images per mask, and selected only four masks, although still evaluating all datasets, totaling 640 images per dataset and 2560 images overall. We used both the FID score, computed using [7], and the LPIPS score, using [9], for each dataset. Unfortunately, the results and conclusions are very similar to the downscaled images for the LPIPS score (4). Our strategy and the finetuned GLIDE model perform similarly, with our model usually scoring a small amount worse. However, on the FID score for the upscaled images (3), we see that on some datasets, namely the ones with more descriptive prompts and images closer to the distribution that GLIDE was trained on, our sampling strategy actually pulls ahead of the fine-tuned GLIDE model on its own, and indeed on average our model performs best. Again, considering that our model uses a reduced version of the RePaint strategy for performance, this is very encouraging and suggests that it may be worth further investigation to determine whether better sampling strategies can improve image quality even on non-inpainting tasks.

	Mean FID Score Per Dataset			
Model	MS-COCO	ImageNet	Places-Train	Mean FID Score
GLIDE-Inpaint	65.65405527	64.25500137	66.55887809	65.48931158
GLIDE-Base w/ RePaint	74.87878045	73.31941275	71.3273807	73.1751913
GLIDE-Inpaint w/ RePaint (ours)	65.45636363	63.81377157	66.87602817	65.38205445

Table 3: Upscaling model performance across different datasets.

	Mean LPIPS Score Per Dataset			
Model	MS-COCO	ImageNet	Places-Train	Mean LPIPS Score
GLIDE-Inpaint	0.330533013	0.345617886	0.346441344	0.340864081
GLIDE-Base w/ RePaint	0.396420178	0.4000803	0.398455561	0.39831868
GLIDE-Inpaint w/ RePaint (ours)	0.347304773	0.35480797	0.349280963	0.350464569

Table 4: Upscaling model performance across different datasets

7.3 Qualitative Results

Figure 4 shows sample results from our method compared to baseline approaches. The non fine-tuned model seems to struggle with the cohesiveness of the image, with colors changing significantly and hard borders showing up around the edge of the mask. This suggests that this model is suffering the most from the limitations in terms of resampling steps that we put on our evaluation. Both the GLIDE-Inpaint finetuned model and our model, which adds the RePaint strategy to this, produce much more cohesive, but still fundamentally flawed images. It is very obvious with any knowledge of the original masks (shown in 5) which mask was used to generate each image, particularly the sr64 image which has a very noticeable screen door effect. Interestingly, both upscaling

models added some depth to the grooves made by this pattern, suggesting some amount of physical realism, and also in some areas the screen door effect was successfully, though inexplicably, removed. In general, our model seems to generate somewhat less pleasant images than the base GLIDE-Inpaint model, though with more of a relationship inferred between the contents of the image and the contents of the prompt. For example, in the stuffed animals image, although GLIDE creates a more pleasant composition, there are no stuffed animals in the image, while our model manages to make the image fit the prompt, and we hypothesize that with further resampling, the overall pleasantness of the image would be improved as well.

7.4 Human Evaluation

Model	GLIDE-Inpaint	GLIDE-Base w/ RePaint	GLIDE-Inpaint w/ RePaint
Average Ranking	1.673076923	2.134615385	1.544871795

Table 5: Average ranking of different models.

We conducted a human evaluation study to assess the perceptual quality of our results. Participants were shown original images along with inpainting results from different methods, and asked to rate them on semantic relevance between the images and the prompt. The participants were blind with regards to which images were generated by which model, and even to whether there was any original image. This meant that for some prompts, the participants actually rated the generated images as more relevant than the original, indicating imperfect captions in the MS-COCO dataset. Results are presented in 5, averaged from a total of 672 ranking votes gathered from 14 participants. More details about the survey are shown in the appendix.

The human evaluation confirmed that our combined approach generally outperforms both standard GLIDE inpainting and RePaint with un-finetuned GLIDE Text2ImUnet model. The un-finetuned model with RePaint consistently performed the worst, creating images that looked bizarre and unnatural, while both fine-tuned models were rated similarly, indicating better performance within our step budget. However, even though our model, GLIDE-Inpaint with RePaint sampling, was only given about 10% of the sampling iterations required by the sampling method outlined in [4], it was consistently rated slightly better than the base finetuned model. This confirms our hypothesis, and our qualitative observations based on the samples in the quantitative evaluation.

8 Conclusions and Future Work

8.1 Key Findings

Our work has revealed the following important findings:

- **A sampling strategy cannot replace a fine-tuned model.** We began our project by hypothesizing that RePaint could be a viable alternative to a model fine-tuned for inpainting, like GLIDE's inpainting model. This is not true.
- **A sampling strategy can augment a fine-tuned model.** Our approach of combining RePaint with the GLIDE-Inpainting model often outperformed standard GLIDE Inpainting.
- **A sampling strategy can give us free diffusion steps.** Interestingly, the resampling steps of RePaint, though using the same GLIDE base model, actually executed faster on the A100. Thus, a RePaint batch with 600 iterations only took about 1.15x as long as 250 diffusion steps, and 900 iterations only just over 2x as long. This extra sampling can be used strategically to improve image quality.

8.2 Limitations

The main limitations of our approach that should be acknowledged are:

- **Increased computational cost** over standard inpainting, particularly for large images.
- **Limited to maximum resolution (256×256)** due to the architecture of the underlying GLIDE models.
- **Performance gap compared to state-of-the-art models** like Stable Diffusion, which operate in latent space.

8.3 Future Work

Based on our findings, there are promising directions for future work:

- **Implement DDIM sampling** [8] for faster inference while maintaining quality.
- **Extend to higher resolutions** to make the approach more practical for real-world applications.
- **Adapt RePaint to latent diffusion models** by developing efficient approaches to operate directly in latent space, addressing the challenges outlined in Section 5.4.

References

[1] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255, 2009.

[2] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Proceedings of the 34th International Conference on Neural Information Processing Systems*, NIPS '20, Red Hook, NY, USA, 2020. Curran Associates Inc.

[3] Tsung-Yi Lin, Michael Maire, Serge J. Belongie, Lubomir D. Bourdev, Ross B. Girshick, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO: common objects in context. *CoRR*, abs/1405.0312, 2014.

[4] Andreas Lugmayr, Martin Danelljan, Andres Romero, Fisher Yu, Radu Timofte, and Luc Van Gool. Repaint: Inpainting using denoising diffusion probabilistic models. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11451–11461, 2022.

[5] Alex Nichol, Prafulla Dhariwal, Aditya Ramesh, Pranav Shyam, Pamela Mishkin, Bob McGrew, Ilya Sutskever, and Mark Chen. GLIDE: towards photorealistic image generation and editing with text-guided diffusion models. *CoRR*, abs/2112.10741, 2021.

[6] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. *CoRR*, abs/2112.10752, 2021.

[7] Maximilian Seitzer. pytorch-fid: FID Score for PyTorch. `https://github.com/mseitzer/pytorch-fid`, August 2020. Version 0.3.0.

[8] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *CoRR*, abs/2010.02502, 2020.

[9] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 586–595, 2018.

[10] Bolei Zhou, Agata Lapedriza, Aditya Khosla, Aude Oliva, and Antonio Torralba. Places: A 10 million image database for scene recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 40(6):1452–1464, 2018.

Prompt: A flip phone sitting next to a laptop computer

Mask: sr64



Prompt: A horse pulling a trolley carriage with a man on it

Mask: genhalf



Prompt: There is a table with stuffed animals

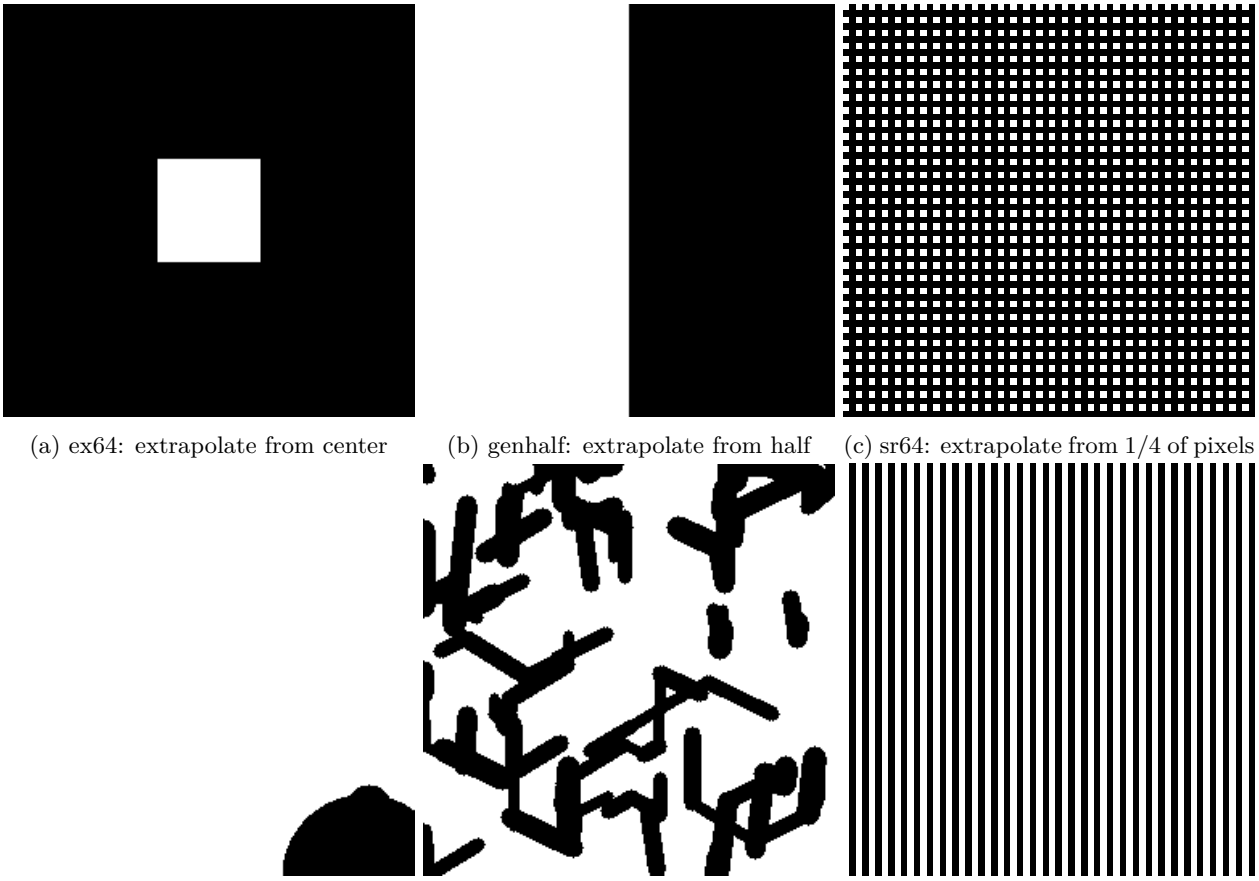
Mask: ex64



Figure 4: Images generated by our method, based on COCO dataset of caption/image pairs. From left to right: Original, RePaint (no Finetuning), Glide Finetuned, RePaint (w/ Glide Finetuned)

Appendix

A Masks



(a) ex64: extrapolate from center (b) genhalf: extrapolate from half (c) sr64: extrapolate from 1/4 of pixels

(d) thick: fill in small blob (e) thin: fill in strokes (f) vs64: fill in every other strip

Figure 5: The masks used in evaluation. Black is masked out, white is kept. Scaled down to 64x64 for the downscaled model.

B Human Evaluation: Survey

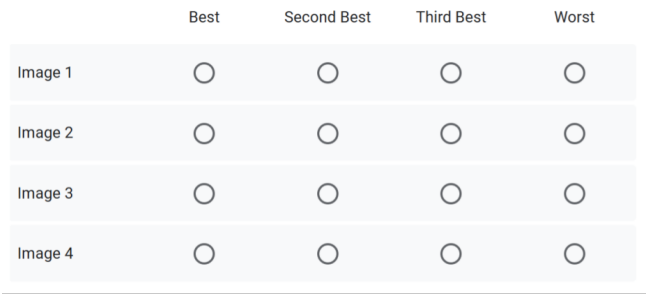


Figure 6: The ranking choices available in the survey for each prompt.



(a) Prompt: A group of young children standing in a room next to each other (mask: ex64)



(b) Prompt: A car and a truck on a city street (mask: ex64)



(c) Prompt: A large chocolate milkshake and a slice of cake (mask: genhalf)



(d) Prompt: A bus is going down the street and has an advertisement on the side of it (mask: genhalf)



(e) Prompt: A man leaning his head in a box of pizza with some pizza in his mouth (mask: sr64)



(f) Prompt: This is an image of a man throwing a frisbee (mask: sr64)



(g) Prompt: An umpire is talking with two players on different teams (mask: thick)



(h) Prompt: An image of a girl getting ready to serve on the tennis court (mask: thick)



(i) Prompt: The teammates are lifting up one of their members (mask: thin)



(j) Prompt: A group of elephants with water in front and trees behind (mask: thin)



(k) Prompt: Some sort of bathroom facility possibly in an airplane (mask: vs64)



(l) Prompt: A man with a baseball bat that is in the dirt (mask: vs64)

Figure 7: The images and prompts used in the human evaluation's survey.

C Complete FID Score Results (64x64) samples

	Per Mask FID Score						
Dataset / Model	ex64	genhalf	sr64	thick	thin	vs64	Overall Mean
COCO / GLIDE-Inpaint	81.99374547	54.01820429	146.6925097	8.620483745	61.86146165	75.97476509	71.52686166
COCO / GLIDE-Base w/ Repaint	87.76922706	65.31123737	158.0123164	15.6368283	107.8254904	118.2560049	92.13518409
COCO / GLIDE-Inpaint w/ Repaint	81.52313851	52.59292502	153.4716241	9.515729039	59.76446022	89.93936977	74.46787444
ImageNet / GLIDE-Inpaint	76.03098924	49.64038824	167.8515051	6.150705451	51.98539905	73.39488325	70.84231172
ImageNet / GLIDE-Base w/ Repaint	80.3127161	63.36554496	165.1959923	9.439924927	102.541331	108.7994535	88.27582714
ImageNet / GLIDE-Inpaint w/ Repaint	76.50100345	50.00625693	163.5905345	6.521871774	48.845419	86.43113157	71.98270286
Places Train / GLIDE-Inpaint	73.94902469	46.45378571	163.3943226	9.535836663	57.9961596	79.64968508	71.82980239
Places Train / GLIDE-Base w/ Repaint	75.01378113	58.79760304	151.1576536	14.70026532	96.22281612	117.5644032	85.57608707
Places Train / GLIDE-Inpaint w/ Repaint	75.89281466	47.70017036	168.1100567	9.167513448	57.80752464	89.14192024	74.63666668

Table 6: Complete FID score table for 64x64 samples

D Evaluation Images

- **Samples 256x256 Paper.tgz**: 256x256 images used for evaluation. 640 images per dataset, 4 datasets. x3 models, + the original image, gives us 10240 images. Google Drive Link
- **Samples 64x64 Paper.tgz**: 64x64 images used for evaluation. 640 images per mask, 6 masks, 4 datasets, 3 models + the original image, gives us 61440 small images. Google Drive Link
- **Samples 256x256 UnFinetuned Upsampler.tgz**: 256x256 images not used for evaluation in this report, but evaluated for presentation. Upscaling the same sampled images from the 64x64 images set, but using the same unconditioned upscaling model on all images.Google Drive Link
- **Samples 64x64 Used to Upscale.tgz**: The images selected from the 64x64 samples for upscaling. It's random, but consistent. Google Drive Link